Assignment → Parameter Estimation.

Q1) as obs are independent, joint density ie likelihood is prod of individual densities.

$$L(\theta_1, \theta_2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\theta_2}} \cdot e^{-\frac{(x-\mu)^2}{2\theta^2}}$$

$$L = (2\pi\theta_2)^{-n/2} \cdot e^{-\frac{1}{2\theta_2}\sum_{i=1}^{n}(x_i-\theta_1)^2}$$

natural log on both sides.

$$\ln L = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\theta_2) - \frac{1}{2\theta_2}\sum_{i=1}^{n}(x_i-\theta_1)^2$$

① Diff wrt θ_1 & set 0.

$$\frac{\partial \ln L}{\partial \theta_1} = \frac{1}{\theta_2}\sum(x_i-\theta_1) = 0$$

$$\sum_{i=1}^{n}(x_i-\theta_1) = 0$$

$$\sum x_i - n\theta_1 = 0$$

$$\boxed{\hat{\theta}_1 = \frac{\sum x_i}{n} = \bar{x}}$$ sample mean.

② Diff w.r.t. θ_2 & set $=0$

$$\frac{\partial \ln L}{\partial \theta_2} = \frac{-n}{2\theta_2} + \frac{1}{2\theta_2^2}\sum_{i=1}^{n}(x_i-\theta_1)^2 = 0$$

multiply by $2\theta_2^2$

$$\rightarrow -n\theta_2 + \sum(x_i-\theta_1)^2 = 0$$

$$\rightarrow \hat{\theta}_2 = \sum_{i=1}^{n}(x_i-\bar{x})^2/n$$ (MLE variance)

Parameter	MLE estimation
θ_1 (mean μ)	$\bar{x} = \sum x_i/n$
θ_2 (variance σ^2)	$\sum(x_i-\bar{x})^2/n$

Q2) for $x \sim B(m, \theta)$: mean : $E(x) = m\theta$

variance : $Var(X) = m\theta(1-\theta)$

let $\hat{\theta}$ be the Estimate of θ we need to express θ in terms of observable data.

from $E(x) = m\theta$ a natural Estimate is -

$$\hat{\theta} = \frac{\bar{x}}{m} = \frac{1}{mn}\sum_{i=1}^{n} x_i$$

$$MSE(\hat{\theta}) = E[(\hat{\theta}-\theta)^2]$$

$$E\left[\left(\theta - \frac{\bar{x}}{m}\right)^2\right]$$

$$E[\hat{\theta}^2] - 2\theta \cdot E[\hat{\theta}] + \theta^2$$

verify $E[\hat{\theta}] = \theta \Rightarrow$

$$E\left[\frac{\bar{x}}{m}\right] = \frac{E[\bar{x}]}{m} = \frac{E[x]}{m} = \frac{m\theta}{m} = \theta$$

Estimator is unbiased $\rightarrow E[\hat{\theta}] = \theta$

Since the Estimator is unbiased MSE = variance of the Estimator

$$MSE(\hat{\theta}) = var\left(\frac{\bar{x}}{m}\right)$$

$$= \frac{1}{m^2} var(\bar{x}) = \frac{1}{m^2} \frac{var(x)}{n}$$

Substituting $Var(X) = m\theta(1-\theta)$

$$= \frac{1}{m^2} \cdot \frac{m\theta(1-\theta)}{n}$$

$$\text{min MSE} = \frac{\theta(1-\theta)}{mn}$$

$\therefore$ MSE Estimate of θ $\hat{\theta} = \frac{\bar{x}}{m} = \frac{\sum x_i}{mn}$

$\therefore$ Minimum MSE $\frac{\theta(1-\theta)}{mn}$